

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$x = \pm \sqrt{-1}$$

$$x = \pm i$$

$$\sqrt{-1} = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

$$\frac{1}{i} = -i$$

$$z = x + iy \rightarrow |z| = \sqrt{x^2 + y^2}$$

$$|z|^2 = x^2 + y^2$$

$$\bar{z} = x - iy$$

$$z + \bar{z} = 2x \rightarrow x = \frac{z + \bar{z}}{2}$$

$$z - \bar{z} = 2iy \rightarrow y = \frac{z - \bar{z}}{2i}$$

$$z \cdot \bar{z} = x^2 + y^2$$

$$z \cdot \bar{z} = |z|^2$$

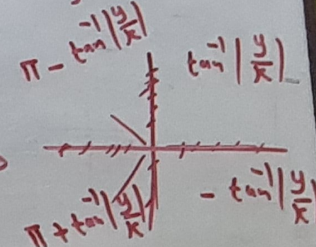
$$\frac{1}{z} = \frac{1}{x+iy} \cdot \frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2}$$

$$z = x + iy = [r, \theta] = r(\cos(\theta) + i\sin(\theta))$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta =$$

$$\begin{aligned} \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \frac{\pi}{6} \\ \tan^{-1}(1) &= \frac{\pi}{4} \\ \tan^{-1}(\sqrt{3}) &= \frac{\pi}{3} \end{aligned}$$

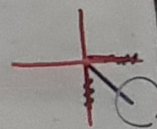


$$\begin{aligned} z &= 2 - 2\sqrt{3}i \quad \begin{matrix} x=2 \\ y=-2\sqrt{3} \end{matrix} \\ r &= \sqrt{(2)^2 + (-2\sqrt{3})^2} = \sqrt{4+12} \\ &= \sqrt{16} = 4 \end{aligned}$$

$$\theta = -\tan^{-1}\left(\frac{2\sqrt{3}}{2}\right)$$

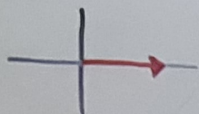
$$\theta = -\tan^{-1}(\sqrt{3})$$

$$\theta = -\frac{\pi}{3}$$



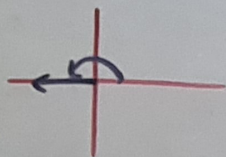
$$z = 3$$

$$r = 3 \quad \theta = 0$$



$$z = -3$$

$$r = 3 \quad \theta = \pi$$



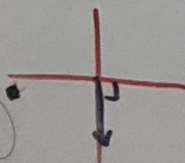
$$z = 3i$$

$$r = 3 \quad \theta = \frac{\pi}{2}$$



$$z = -3i$$

$$r = 3 \quad \theta = -\frac{\pi}{2}$$



$$z = x + iy$$

$$= (r, \theta) = r(\cos(\theta) + i\sin(\theta))$$

$$= r e^{i\theta}$$

$$x + iy = r(\cos \theta + i\sin \theta) = r e^{i\theta}$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$e^{-i\theta} = \cos(\theta) - i\sin(\theta)$$

$$\ln(z) = \ln(r e^{i\theta})$$

$$= \ln(r) + \ln(e^{i\theta})$$

$$\ln(z) = \ln(r) + i\theta$$

$$= \ln(\sqrt{x^2 + y^2}) + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \ln(x^2 + y^2)^{\frac{1}{2}} + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \frac{1}{2} \ln(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$|z| = 3$$

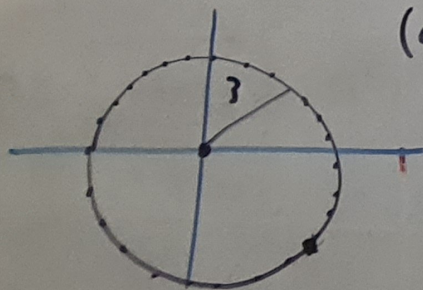
$$|x + iy| = \sqrt{x^2 + y^2}$$

$$\sqrt{x^2 + y^2} = 3$$

$$x^2 + y^2 = 9$$

$$r = 3$$

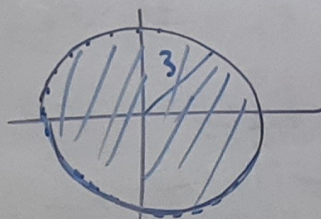
$$(0, 0)$$



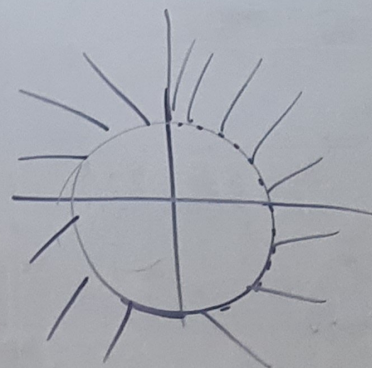
$$|z| \leq 3$$

$$\sqrt{x^2 + y^2} \leq 3$$

$$x^2 + y^2 \leq 9$$



$$|z| \geq 3$$



$$|z - 3| = 5$$

$$(3, 0) \quad r=5$$

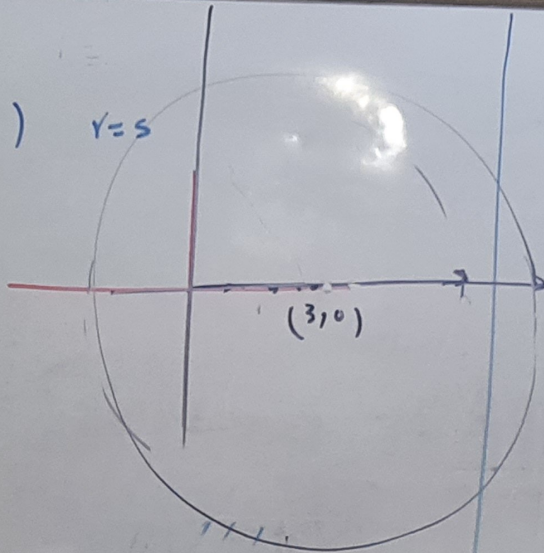
$$|x+iy - 3| = 5$$

$$|(x-3) + iy| = 5$$

$$\sqrt{(x-3)^2 + y^2} = 5$$

$$(x-3)^2 + y^2 = 25$$

$$(3, 0) \quad r=5$$



$$|z + 2| = 3$$

$$r=3 \quad (-2, 0)$$

$$|z - 3i| = 4$$

$$r=4 \quad (0, 3)$$

$$|x+iy - 3i| = 4$$

$$|x + (y-3)i| = 4$$

$$\sqrt{x^2 + (y-3)^2} = 4$$

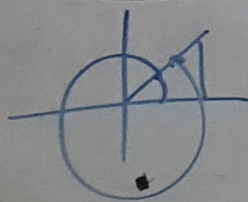
$$\sqrt[n]{z} = (x+iy)^{\frac{1}{n}}$$

$$= (r e^{i\theta})^{\frac{1}{n}} = r^{\frac{1}{n}} e^{i\frac{\theta}{n}}$$

$$= r^{\frac{1}{n}} [\cos(\frac{\theta}{n}) + i \sin(\frac{\theta}{n})]$$

$$\sin(\theta) = \sin(\theta + 2\pi k)$$

$$\cos(\theta) = \cos(\theta + 2\pi k)$$



$$k = 0, 1, 2, \dots$$

$$\sqrt[n]{z} = (x+iy)^{\frac{1}{n}} = [r e^{i\theta}]^{\frac{1}{n}}$$

$$= [r e^{i(\theta + 2\pi k)}]^{\frac{1}{n}}$$

$$= r^{\frac{1}{n}} e^{i \frac{(\theta + 2\pi k)}{n}}$$

$$= r^{\frac{1}{n}} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right]$$

$$k = 0, 1, 2, 3, \dots$$

$$\sqrt[n]{z} = e^{c \ln(z)}$$

$$= e^{c(\ln(r) + i\theta)}$$

$$\ln(z) = \ln(r e^{i\theta})$$

$$= \ln(r) + \ln e^{i\theta}$$

$$= \ln(r) + i\theta$$

$$y = f(x)$$

دالة حقيقية

$$y = x^2 + 2x - 1$$

$$f(x) = x^2 + 2x - 1$$

$$w = f(z)$$

دالة مركبة

$$w = z^2 - 2z + 1$$

$$f(z) = z^2 - 2z + 1$$

$$f(z) = f(x+iy) = \underline{u(x,y)} + i \underline{v(x,y)}$$

$$f(z) = z^2$$

$$= (x+iy)^2$$

$$= x^2 + 2ixy + i^2 y^2$$

$$= x^2 + 2ixy - y^2$$

$$= \underbrace{(x^2 - y^2)}_u + \underbrace{2ixy}_v$$

$$u = x^2 - y^2$$

$$v = 2xy$$

$$f(z) = |z|^2$$

$$= (\sqrt{x^2 + y^2})^2$$

$$= x^2 + y^2$$

$$u = x^2 + y^2$$

$$v = 0$$

$$y = f(x)$$

دالة حقيقية

$$y = x^2 + 2x - 1$$

$$f(x) = x^2 + 2x - 1$$

$$w = f(z)$$

دالة مركبة

$$w = z^2 - 2z + 1$$

$$f(z) = z^2 - 2z + 1$$

$$f(z) = f(re^{i\theta}) = u(r, \theta) + iv(r, \theta)$$

$$f(z) = z + \frac{1}{z}$$

$$= r e^{i\theta} + \frac{1}{r e^{i\theta}} = r e^{i\theta} + \frac{1}{r} e^{-i\theta}$$

$$= r [\cos(\theta) + i \sin(\theta)] + \frac{1}{r} [\cos(\theta) - i \sin(\theta)]$$

$$= r \cos(\theta) + r i \sin(\theta) + \frac{1}{r} \cos(\theta) - \frac{1}{r} i \sin(\theta)$$

$$= \underbrace{\left(r + \frac{1}{r}\right) \cos \theta}_u + \underbrace{\left(r - \frac{1}{r}\right) \sin \theta}_v i$$

$$y = f(x)$$

دالة حقيقية

$$y = x^2 + 2x - 1$$

$$f(x) = x^2 + 2x - 1$$

$$w = f(z)$$

دالة مركبة

$$w = z^2 - 2z + 1$$

$$f(z) = z^2 - 2z + 1$$

$$\sinh(\theta) = \frac{e^{\theta} - e^{-\theta}}{2}$$

$$\cosh(\theta) = \frac{e^{\theta} + e^{-\theta}}{2}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin(-\theta) = -\sin(\theta)$$

$$\cos(-\theta) = \cos(\theta)$$

$$\tan(-\theta) = -\tan(\theta)$$

$$\sin(i\theta) = i \sinh(\theta)$$

$$\cos(i\theta) = \cosh(\theta)$$

$$\tan(i\theta) = i \tanh(\theta)$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta) \rightarrow 1$$

$$e^{-i\theta} = \cos(\theta) - i\sin(\theta) \rightarrow 2$$

$$2\cos(\theta) = e^{i\theta} + e^{-i\theta}$$

$$2i\sin(\theta) = e^{i\theta} - e^{-i\theta}$$

$$y = f(x)$$

دالة حقيقية

$$y = x^2 + 2x - 1$$

$$f(x) = x^2 + 2x - 1$$

$$w = f(z)$$

دالة مركبة

$$w = z^2 - 2z + 1$$

$$f(z) = z^2 - 2z + 1$$

$$\sinh(\theta) = \frac{e^{\theta} - e^{-\theta}}{2}$$

$$\cosh(\theta) = \frac{e^{\theta} + e^{-\theta}}{2}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin(-\theta) = -\sin(\theta)$$

$$\cos(-\theta) = \cos(\theta)$$

$$\tan(-\theta) = -\tan(\theta)$$

$$\sin(i\theta) = i \sinh(\theta)$$

$$\cos(i\theta) = \cosh(\theta)$$

$$\tan(i\theta) = i \tanh(\theta)$$

$$\sinh(i\theta) = i \sin(\theta)$$

$$\cosh(i\theta) = \cos(\theta)$$

$$\tanh(i\theta) = i \tan(\theta)$$

$$\cos(i\theta) = \frac{e^{i(i\theta)} + e^{-i(i\theta)}}{2}$$

$$= \frac{e^{-\theta} + e^{\theta}}{2}$$

$$= \frac{e^{\theta} + e^{-\theta}}{2}$$

$$= \cosh(\theta)$$